Fluid and Bead Parameters

Hydrophobic spheres floating on a liquid interface: inputs and computed contact radii · November 2, 2026

1. Fluid

Quantity	Symbol	Value
Density	ρ_f	1148.0 kg/m^3 (1.148 g/cm^3)
Surface tension	γ	0.068 N/m (68 mN/m)
Gravitational acceleration	g	9.81 m/s^2
Air density	ρ_a	neglected (taken as 0)
Capillary length (derived)	$\ell_c = (\gamma / \rho_f g)^{1/2}$	2.457 mm

2. Beads

All beads carry the same hydrophobic coating. A contact angle of **θ = 160°** is assumed for every bead (not measured).

Bead	Source / part no.	Material	Diameter	Radius R	Density ρ_s
PTFE (reference)	Reference bead from Thomson et al. (2026)	PTFE	1.5800 mm	0.7900 mm	2.20 g/cm^3
Aluminum 1/32"	McMaster-Carr 34665K21	2017 aluminum	1/32" (0.7938 mm)	0.3969 mm	2.79 g/cm^3
Aluminum 1/16"	McMaster-Carr 34665K22	2017 aluminum	1/16" (1.5875 mm)	0.7938 mm	2.79 g/cm^3
Nylon 1/16"	McMaster-Carr 9613K11	Wear-resistant nylon (nylon 6/6)	1/16" (1.5875 mm)	0.7938 mm	1.14 g/cm^3

3. Computed floating position and contact radius

From the full nonlinear axisymmetric Laplace–Young equation coupled to the vertical force balance on the sphere (Thomson, Barotta & Harris, *Phys. Rev. Research* **8**, L022021 (2026), Supplemental Material §2.2.2). ψ is the angular position of the contact line, measured from the bottom of the sphere; the contact radius is $r_c = R \sin\psi$.

Bead	Bond number Bo = $(R/\ell_c)^2$	Density ratio D = ρ_s/ρ_f	ψ	Contact radius r_c	r_c / R
PTFE (reference)	0.1034	1.916	33.34°	**0.4342 mm**	0.5497
Aluminum 1/32"	0.0261	2.430	25.57°	**0.1713 mm**	0.4316
Aluminum 1/16"	0.1043	2.430	35.98°	**0.4663 mm**	0.5875
Nylon 1/16"	0.1043	0.993	28.14°	**0.3743 mm**	0.4716

4. Assumptions and notes

- The aluminum and nylon densities are nominal handbook values for the bulk materials (2017 aluminum 2.79 g/cm^3; nylon 6/6 1.14 g/cm^3), since the supplier does not list them. Weighing a batch of beads would pin these down.
- The contact angle θ = 160° is assumed for the coated beads; the mass of the coating is neglected.
- The nylon beads are very slightly less dense than the fluid (D ≈ 0.99), so they would float on buoyancy alone; the other beads are denser than the fluid and are held at the surface by surface tension.
- All beads are treated as isolated, perfect spheres on an otherwise flat, quiescent interface.